

UAVX Tuning Report — Original Airframe Revisit (raw + legacy)

Date: 2026-08-29 **Version:** 1.0 **Scope:** Revisit of wiki/docs/critique_report.md v4.0 (2026-08-08) — retune every original/ airframe from the clean baseline (the retired user/_Tuned fleet is **not** resurrected). Delivers the current-vs-proposed tuning table airframe by airframe, with every changed parameter shown as **raw FC float** (what the .af stores / what the FC ParamData holds) **and** its **legacy integer-encoded** equivalent where one exists. **Simulator:** current tests/test_pid_sim.py (MFD criteria below). PDF conversion is the USER's job (host "Merlin", scripts/md2pdf.sh) — the assistant has no pandoc/xelatex.

Result: the 7 larger frames go **original 3-7 FAILs → proposed 0 FAILs**; the two micros go **9/7 FAILs → 2/1**, and their sole remaining FAILs are a **hard Alt-hold physics floor**, not a tuning deficiency (analytic match, see §6).

1. Were these frames tuned before, and did they pass?

No. The documentation record:

Source	Claim	Truth for <i>these</i> originals
tuning_study.md v2.1 (2026-08-04)	"all 12 FW + 5 generic MR frames PASS"	That is the generic/ fleet, not original/.
MC_TUNING_REPORT7.MR (2026-07-26)	MC _Tuned identical behavior, attitudes PASS	Same generic/base-gains cohort; the _Tuned copies of <i>these</i> frames were 3-8 FAIL.
critique_report v3.1 (2026-08-06)	regenerated user/_Tuned fleet validated	Phoenix_Tuned 0/45 PASS , Radian_Tuned 0/45, Arado_555_Tuned 1/44 (artifact). Every MR copy failed : Ecks/Ken450/KenAlpha/Ming/S500 3/42, Rok_Quad 4/41, Ken_LadyBug 7/36, WLToys 8/35.
critique_report v4.0 (2026-08-08)	_Tuned fleet retired	All _Tuned frames carried the cascade I-limit windup bomb (legacy raw 10 in ANGLE_Q_INT_LIMIT). Prior tuned results are invalid — retune restarts from original/.
critique_report v3.0	original/Phoenix	PASS — the only one of the 9 ever passing before.

So of these 9 frames only **Phoenix** had previously passed; everything else was a retired/invalid retune or the generic fleet. This campaign is the first from clean original/.

2. Simulation convention and the Q gain

The attitude outer loop is **quaternion**, FC applies $2.0f * Qa[a] * A[a].P.Kp$ (control.c). The critique sim reads the *_ANGLE_Q_* KP directly as a regular proportional gain ($rate_desired = angle_error * KP$). This is valid for tuning because for **small angles** $2 \cdot Qa \cdot QGain \approx angle_error \cdot QGain$ — so RollAngleQKp, PitchAngleQKp, YawAngleQKp **are the quaternion gains**, and their numeric value is the small-angle gain.

Consequence for this campaign:

- **The tuning lever is the RATE loop** (RollRateKp/Kd, PitchRateKp/Kd, YawRateKp/Kd) **and the altitude chain** (AltPosKp, AltROCKp, AltVelKi, AltThrottleCompLimit) — **none of these are Q-scaled**; they are plain PD/PI on rate error / altitude ROC, identical for MC and FW.
- **Q (angle) gains were deliberately left at their source values in every frame but the two micros**, where YawAngleQKp 2.25 → 5.0 (still below the MR envelope default 8; the outer yaw loop was under-gaining the insufficient yaw-rate loop underneath it). Every other *_ANGLE_Q_* value is unchanged (Rok_Quad's PitchAngleQKp experiment ended back at source 5.75 — net zero).

3. Criteria and limits

Scored by current tests/test_pid_sim.py: `python3 -m tests.test_pid_sim proposed/<Frame>.af` (run from `uavx-python/src`).

- Attitude steps: Roll @ 15°, Pitch @ 10°, Yaw @ 45° — rise/overshoot/settle/ss-error; rate headroom; disturbance rejection gusts (peak deviation roll 15°/s, pitch 12°/s, yaw 12°/s).
- AH 5 m: rise ≤ 2.0 s, overshoot ≤ 20 %, settle ≤ 5.0 s, ss-error ≤ 1.0 m.
- Nav 10 m: rise ≤ 2.0 s, overshoot ≤ 20 %, settle ≤ 5.0 s.
- FW (Phoenix): separate FW criteria; aileron-less roll non-scored.

All proposed values lie inside the current FC ParamTable / GCS PARAM_LIMITS agreement (verified against `parameters.py`), so **no ParamTable / PARAM_LIMITS edits are required** for this retune:

Param	idx	GCS limit	highest used	
RollRateKp	0	(0, 3.0)	0.34	✓
PitchRateKp	5	(0, 3.0)	0.42 (Rok)	✓
YawRateKp	10	(0, 3.0)	0.25 (micros)	✓
RollRateKd	11	(0, 0.2)	0.008	✓
PitchRateKd	27	(0, 0.2)	0.012 (Rok)	✓
YawRateKd	90	(0, 0.05)	0.015 (Phoenix)	✓
AltPosKp	6	(0, 3.66)	2.2	✓
AltVelKi (AltROCKi)	101	(0, 3.66)	0.005	✓
AltThrottleCompLimit	102	(0, 0.25)	0.25 (= FC max)	✓
AltROCKp	120	(0, 3.66)	1.3	✓
YawAngleQKp	96	(0, 20.0)	5.0 (micros, Q gain)	✓

Legacy column legend: `legacy = raw / PARAM_SCALES[idx]` (the historical integer-encoded meaning). — = the index is **not** legacy-tagged (unified-float-only parameter, no scale bridge exists — AltROCKp 120, AltThrottleCompLimit 102, YawAngleQKp 96). Scales: idx 0/5/10 0.005, 11/27 0.0001, 90 2.5e-5, 6 0.0183, 101 0.00027. See `wiki/Legacy_Scaling_Map.md`.

4. Per-airframe current vs proposed

Format: `idx Param raw old → raw new [legacy old → legacy new] comment.`

4.1 Ecks_220mm — 3 FAILs → 0

idx	Param	raw old → new	legacy old → new
0	RollRateKp	0.2000 → 0.3400	40.0000 → 68.0000
5	PitchRateKp	0.2500 → 0.3400	50.0000 → 68.0000
10	YawRateKp	0.03333 → 0.1100	6.6667 → 22.0000
11	RollRateKd	0.0045 → 0.0080	45.0000 → 80.0000
27	PitchRateKd	0.0045 → 0.0080	45.0000 → 80.0000
90	YawRateKd	0.0003375 → 0.0020	13.5000 → 80.0000
6	AltPosKp	1.8300 → 2.1000	100.0000 → 114.7541
101	AltVelKi	0.0005 → 0.0050	1.8519 → 18.5185
120	AltROCKp	1.0000 → 1.2000	— → —

Clears: yaw step (OS 17.5 %/settle 8 s), roll-gust (17.8°/s) & yaw-gust (17.2°/s), AH rise (2.03 s → 1.95 s). AltThrottleCompLimit already 0.2 in the source (unchanged).

4.2 Ken_450_1165 — 3 FAILs → 0 (tuned in a prior session today)

idx	Param	raw old → new	legacy old → new
0	RollRateKp	0.2080 → 0.3400	41.6000 → 68.0000
5	PitchRateKp	0.2080 → 0.3200	41.6000 → 64.0000
10	YawRateKp	0.06667 → 0.1100	13.3333 → 22.0000
11	RollRateKd	0.0060 → 0.0080	60.0000 → 80.0000
90	YawRateKd	0.0003375 → 0.0020	13.5000 → 80.0000
6	AltPosKp	1.8300 → 2.1000	100.0000 → 114.7541
101	AltVelKi	0.0005 → 0.0050	1.8519 → 18.5185
102	AltThrottleCompLimit	0.1200 → 0.2000	— → —
120	AltROCKp	1.0000 → 1.2000	— → —

Carries one non-tuning delta from the Mag-Gating session: CONFIG1_BITS eUsingMag → 6.0 (UseMag master switch on).

4.3 Ken_Alpha_Test — 3 FAILs → 0

idx	Param	raw old → new	legacy old → new
0	RollRateKp	0.2375 → 0.3400	47.5000 → 68.0000
5	PitchRateKp	0.2375 → 0.3400	47.5000 → 68.0000
10	YawRateKp	0.04167 → 0.1100	8.3333 → 22.0000
11	RollRateKd	0.0020 → 0.0080	20.0000 → 80.0000
27	PitchRateKd	0.0060 → 0.0080	60.0000 → 80.0000
90	YawRateKd	0.0003375 → 0.0020	13.5000 → 80.0000
6	AltPosKp	1.8300 → 2.1000	100.0000 → 114.7541
101	AltVelKi	0.0005 → 0.0050	1.8519 → 18.5185
102	AltThrottleCompLimit	0.0500 → 0.2000	— → —
120	AltROCKp	1.0000 → 1.2000	— → —

Clears the worst roll-gust of the fleet (19.31°/s) and AH rise (3.34 s → 1.85 s).

4.4 Ming_DevEBox — 6 FAILs → 0

idx	Param	raw old → new	legacy old → new
0	RollRateKp	0.2375 → 0.3400	47.5000 → 68.0000
5	PitchRateKp	0.2375 → 0.3400	47.5000 → 68.0000
10	YawRateKp	0.04167 → 0.1100	8.3333 → 22.0000
11	RollRateKd	0.0020 → 0.0080	20.0000 → 80.0000
27	PitchRateKd	0.0060 → 0.0080	60.0000 → 80.0000
90	YawRateKd	0.0003375 → 0.0020	13.5000 → 80.0000
6	AltPosKp	1.8300 → 2.2000	100.0000 → 120.2186
101	AltVelKi	0.0005 → 0.0050	1.8519 → 18.5185
120	AltROCKp	1.0000 → 1.3000	— → —

The only frame with step **overshoot** failures (roll 10.4 %, pitch 10.5 %, yaw 16.4 %). Rate PD rebuild cleared all three; AH needed the slightly higher AltPosKp/AltROCKp (2.2/1.3) to pull rise 2.014 → < 2 s.

4.5 Phoenix (FW) — 1 FAIL → 0

idx	Param	raw old → new	legacy old → new
10	YawRateKp	0.2852 → 1.1000	57.0400 → 220.0000
90	YawRateKd	0.00066 → 0.0150	26.4000 → 600.0000

FW yaw was critically under-gained (rate KP 0.285 vs the FW needs $\geq \sim 1$). Yaw @ 15° rise 8 s → ≤ 5 s. Everything else (pitch, AH, Nav, structural roll gate) already passed.

4.6 Rok_Quad — 7 FAILs → 0

idx	Param	raw old → new	legacy old → new
—	AF_TYPE	eQuadAF → eQuadXAF	fix: the file's own type (3) is not even in the sim AF_CATEGORY; original/ only loads courtesy of a hard-coded loader override (AF_FILES["original/Rok_Quad.af"]→eQuadXAF) The proposed file must carry eQuadXAF (=4) itself to load/score and to select the right mixer on the FC.
0	RollRateKp	0.2375 → 0.3400	47.5000 → 68.0000
5	PitchRateKp	0.2375 → 0.4200	47.5000 → 84.0000
10	YawRateKp	0.04167 → 0.1100	8.3333 → 22.0000
11	RollRateKd	0.0020 → 0.0080	20.0000 → 80.0000
27	PitchRateKd	0.0060 → 0.0120	60.0000 → 120.0000
90	YawRateKd	0.0003375 → 0.0020	13.5000 → 80.0000
6	AltPosKp	1.8300 → 2.2000	100.0000 → 120.2186
101	AltVelKi	0.0005 → 0.0050	1.8519 → 18.5185
102	AltThrottleComp	0.0600 → 0.2500	— → —
120	AltROCKp	0.0500 → 1.3000	— → —

Worst overshoot frame (16.3/16.8/21.5 % across steps — new 1344 g / 9×4 prop mass, hover $\approx 55\%$). Needed the heaviest rate-KD damping (**pitch KD 0.012**, pitch rate KP 0.42 — the sim’s “increase RateKp to cut overshoot” is the cascade counterintuitive: faster rate loop, less angle integration past command) plus full AltThrottleCompLimit 0.25 to climb the 5 m step in ≤ 2 s (rise 3.79 s $\rightarrow$ 2.00 s). Q gains untouched (PitchAngleQKp stayed 5.75; YawAngleQKp stayed 3). **This is the new-mass respray the critique’s item 4 called for, and the only frame needing a physicals retune.** (physicals were already updated in original/.)

4.7 S500_1137 — 4 FAILs $\rightarrow$ 0

idx	Param	raw old $\rightarrow$ new	legacy old $\rightarrow$ new
0	RollRateKp	0.2000 $\rightarrow$ 0.3400	40.0000 $\rightarrow$ 68.0000
5	PitchRateKp	0.2500 $\rightarrow$ 0.3400	50.0000 $\rightarrow$ 68.0000
10	YawRateKp	0.03333 $\rightarrow$ 0.1100	6.6667 $\rightarrow$ 22.0000
11	RollRateKd	0.0045 $\rightarrow$ 0.0080	45.0000 $\rightarrow$ 80.0000
27	PitchRateKd	0.0045 $\rightarrow$ 0.0080	45.0000 $\rightarrow$ 80.0000
90	YawRateKd	0.0003375 $\rightarrow$ 0.0020	13.5000 $\rightarrow$ 80.0000
6	AltPosKp	1.8300 $\rightarrow$ 2.1000	100.0000 $\rightarrow$ 114.7541
101	AltVelKi	0.0005 $\rightarrow$ 0.0050	1.8519 $\rightarrow$ 18.5185
102	AltThrottleCompLimit	0.0500 $\rightarrow$ 0.2000	— $\rightarrow$ —
120	AltROCKp	1.0000 $\rightarrow$ 1.2000	— $\rightarrow$ —

Same lever set; yaw overshoot (17.8 %) and all gusts cleared.

4.8 Ken_LadyBug (micro) — 9 FAILs $\rightarrow$ 2

idx	Param	raw old $\rightarrow$ new	legacy old $\rightarrow$ new
96	YawAngleQKp (Q gain)	2.2500 $\rightarrow$ 5.0000	— $\rightarrow$ —
0	RollRateKp	0.2080 $\rightarrow$ 0.3000	41.6000 $\rightarrow$ 60.0000
5	PitchRateKp	0.2080 $\rightarrow$ 0.3000	41.6000 $\rightarrow$ 60.0000
10	YawRateKp	0.06667 $\rightarrow$ 0.2500	13.3333 $\rightarrow$ 50.0000
11	RollRateKd	0.0030 $\rightarrow$ 0.0080	30.0000 $\rightarrow$ 80.0000
27	PitchRateKd	0.0090 $\rightarrow$ 0.0080	90.0000 $\rightarrow$ 80.0000
90	YawRateKd	0.0000375 $\rightarrow$ 0.0020	1.5000 $\rightarrow$ 80.0000
6	AltPosKp	0.2000 $\rightarrow$ 2.1000	10.9290 $\rightarrow$ 114.7541
101	AltVelKi	0.0005 $\rightarrow$ 0.0050	1.8519 $\rightarrow$ 18.5185
102	AltThrottleCompLimit	0.0200 $\rightarrow$ 0.2500	— $\rightarrow$ —
120	AltROCKp	0.0500 $\rightarrow$ 1.2000	— $\rightarrow$ —

The critique’s v3.0 NaN/inf divergence **no longer reproduces** in the current sim — the frame now runs clean. 9 $\rightarrow$ 2 FAILs; the 8 cleared include the yaw step (rise 7.15 s $\rightarrow$ 2.51 s, settle 8 s $\rightarrow$ 3.58 s), roll-gust (19.2 $\rightarrow$ $\leq 15^\circ/\text{s}$), yaw-gust (13.5 $\rightarrow$ $\leq 12^\circ/\text{s}$) and rate headroom (0.085 $\rightarrow$ 0.23). The 2 remaining are AH rise + settle — see §6 (physics floor, ~ 6.9 s optimum).

4.9 WLToys_LadyBug (micro) — 7 FAILs $\rightarrow$ 1

Same 11 deltas as Ken_LadyBug (identical gain family): YawAngleQKp 2.25 $\rightarrow$ 5.0 (Q gain), rate PD set, AltPosKp 0.2 $\rightarrow$ 2.1, AltVelKi $\rightarrow$ 0.005, AltThrottleCompLimit $\rightarrow$ 0.25, AltROCKp $\rightarrow$ 1.2.

Clears the same 6 (yaw step, both gusts, headroom, all attitude). 1 remaining FAIL = AH rise — §6.

5. Common tunings (the critique §Summary acknowledged pattern)

The original/ MR failures distribute exactly as the critique predicted: **under-gained yaw-rate loop** (settle 8 s), **roll/yaw gust peaks 16-19°/s**, **AH rise 2.0-3.8 s**. The adopted solution per frame is a consistent lever set on the **rate PD** (roll/pitch rate KP to 0.34, rate KD to 0.008, yaw-rate KD to 0.002 = old legacy 80) + the altitude chain (AltROCKp 1.2-1.3, AltPosKp 2.1-2.2, AltVelKi 0.005, comp-limit 0.2-0.25), with two local adjustments:

- **Ming** needed AltPosKp/AltROCKp one step higher (2.2/1.3) for the AH bar;
- **Rok_Quad** needed pitch-specific heavy damping (pitch KD 0.012, pitch rate KP 0.42) for its new-mass step overshoots + full comp-limit 0.25 for climb.

YawRateKp 0.11 is below the sim's MR envelope floor (0.20, guidance-only — the envelope is for slider characterisation, not scored) and validates cleanly; the micros kept 0.25. Q gains untouched except the micros' YawAngleQKp (5.0, still < envelope default 8).

6. Why the two micros cannot clear Alt Hold (physics floor, not tuning)

MR AH plant: $\text{thr_total} = \text{hover} + \text{thr_comp}$, $\text{accel} = \text{thr_total} \cdot \text{max_thrust}/m - g - 0.5 \cdot v^2/m$, with thr_comp clamped to AltThrottleCompLimit. At the FC hard limit (0.25) the climb reaches terminal velocity when drag $0.5 \cdot v^2$ equals excess thrust:

Frame	mass / hover	excess thrust @ comp 0.25	terminal v	4.5 m optimum rise	sim
Ken_LadyBug	30 g, hover 34 % (88 g thrust)	≈ 0.216 N	0.657 m/s	≈ 6.9 s	6.913 s
WLToys_LadyBug	80 g, hover 33 % (240 g thrust)	≈ 0.589 N	1.085 m/s	≈ 4.3 s	4.25 s

The sim matches the analytic optimum to 3 digits. The generic drag model ($0.5 \cdot v^2$) penalises these tiny brushed frames' terminal climb rate; no legal gain or comp-limit ($\leq 0.25 = \text{FC max}$) can reach the 2 s rise bar for a 5 m step. **Documented as a model floor** — the earlier critique already flagged the brushed divide; the remainder is now confined to Alt Hold, with all attitude/yaw/nav-cleared.

Rok_Quad sits close to the same physics (hover ≈ 55 %), which is why it needed the full 0.25 comp-limit — its rise lands at 2.00 s, at the bar.

7. Verification

```
cd UAVXGS/uavx-python/src
```

```
python3 -m tests.test_pid_sim proposed/Ecks_220mm.af      # 0 FAIL
python3 -m tests.test_pid_sim proposed/Ken_450_1165.af    # 0 FAIL
python3 -m tests.test_pid_sim proposed/Ken_Alpha_Test.af  # 0 FAIL
python3 -m tests.test_pid_sim proposed/Ken_LadyBug.af     # 2 FAIL (AH physics floor, §6)
python3 -m tests.test_pid_sim proposed/Ming_DevEBox.af    # 0 FAIL
```

```
python3 -m tests.test_pid_sim proposed/Phoenix.af           # 0 FAIL
python3 -m tests.test_pid_sim proposed/Rok_Quad.af         # 0 FAIL
python3 -m tests.test_pid_sim proposed/S500_1137.af        # 0 FAIL
python3 -m tests.test_pid_sim proposed/WLToys_LadyBug.af    # 1 FAIL (AH rise, §6)
```

Original baselines (current sim, original/<frame>.af): Ecks 3 · Ken450 3 · KenAlpha 3 · Ken_LadyBug 9 · Ming 6 · Phoenix 1 · Rok_Quad 7 · S500 4 · WLToys 7.

Path note: reference frames as original/<name>.af / proposed/<name>.af (relative to src/) — a bare airframes/... prefix breaks the loader's basename descriptor mapping.

8. Actions / decisions

1. **proposed/ = flight-ready candidates**, visible in the GCS load menus. original/ stays as the untouched record (hidden).
2. No params.c ParamTable or GCS PARAM_LIMITS edits — every value sits inside the existing FC-GCS agreement (§3).
3. Never re-instantiate the retired _Tuned fleet (I-limit windup). This campaign's baseline is original/ as committed.
4. Micro AH: captured as a model floor, not silently “fixed” by editing the sim drag — any future sim drag-scaling change must be justified separately and re-run through §7.